District Health Infrastructure Allocation System

AN END-TO-END ANALYTICS PROJECT ON INDIAN PUBLIC HEALTH DATA

Where should a State Health Society build its next 25 rural Sub-Centres — and is an optimised answer measurably better than a sorted list?

705
districts reconciled

696
districts scored

10,000
weight vectors tested

76
automated tests

PostgreSQL · Python · Integer programming (CBC) · Streamlit

Public data: NFHS-5 (2019-21) · Census of India 2011 · IPHS 2022

Portfolio project. Not a policy recommendation.

What this project is

India delivers rural public health through a three-tier structure of Sub-Centres, Primary Health Centres and Community Health Centres. State Health Societies allocate a fixed annual budget of new facilities. In practice those decisions lean on population counts and political considerations, even though district-level health *outcome* data is public.

This system takes that public data and answers a single, concrete question:

THE QUESTION

A State Health Society has sanctioned budget for **25 new Sub-Centres**. Given district-level health outcome data, which districts should receive them — and how much measurable improvement does the optimal allocation deliver over a naive one?

It is a complete pipeline, not a notebook: public data in, a reproducible database, a documented scoring model, a constrained optimiser, a robustness analysis, and an interactive app a non-technical user can drive.

The answer

25 Sub-Centres allocated across 696 eligible rural districts:

Measure	Result
States receiving facilities	11
Regions covered	6 of 6
Terrain mix	16 plains, 5 hilly, 4 tribal
Robust under plausible re-weighting	11 of 25
Contested (depend on the weights)	14 of 25
Never reproducible (excluded)	0 of 25

The ranking is led by Araria, Purnia and Sitamarhi in Bihar's Seemanchal region and Pakur, Sahibganj and Deoghar in Jharkhand's Santhal Pargana — among the most deprived districts in India. The index found them from seven survey indicators with no geographic input and no tuning toward an expected answer.

The claim we actually make is deliberately narrow:

> *“11 of 25 districts survive any plausible re-weighting. 14 depend on what you decided to value. Here is exactly which is which.”*

How it works

Eight stages, each one runnable and independently verifiable.

Stage	What happens
0 · Acquire	Download three public sources, checksum them, assert the shape of each file before anything else runs
1 · Reconcile	Match 2019-21 districts to 2011 Census districts across a decade of splits, renames and new states
1 · Load	Two-schema PostgreSQL model: verbatim staging, typed and constrained core
2 · Score	Composite Need Index built as a chain of SQL views, every stage inspectable
3 · Allocate	Integer program solved to a proven optimum, against three baselines
4 · Stress-test	10,000 alternative weightings; the whole optimisation re-solved 400 times
5 · Map & app	Polygon crosswalk and an interactive Streamlit tool
6 · Document	Six generated reports plus a build log recording every failure

Scoring a district

1. Direction-normalise. Four of the seven indicators are good things (institutional births, antenatal visits, skilled attendance, immunisation) and three are bad (stunting, anaemia). Good ones are inverted so that higher always means greater need.

2. Percentile-rank nationally. Institutional births spans roughly 20-100; stunting spans 6-60. Averaging raw values would let the widest-spread indicator dominate *by accident*, while the weight vector claimed all seven were equal. Ranking puts them on one scale, so the weights are the only thing that decides influence.

3. Weighted mean over the indicators a district actually has, with weights renormalised. No value is ever imputed from another district — an imputed need score driving a real budget allocation is not defensible.

The optimisation

$$\text{maximise} \quad \text{SUM need}_d \times \min(1, \text{rural\_pop}_d / \text{catchment\_norm}_d) \times x_d$$

$$\text{subject to} \quad \text{SUM } x_d = 25 \quad \text{(budget, exactly)}$$

$$\text{SUM}_{\text{state}}\ x_d \leq 4 \quad \text{(equity)}$$

$$\text{SUM}_{\text{region}}\ x_d \geq 1 \quad \text{(political feasibility)}$$

Solved by CBC to a *proven* optimum — not a good-enough answer.

Three findings

1 · The supply adjustment cancels algebraically

The methodology asks for underservice = population at risk divided by existing facilities, falling back to *norm-implied* facilities where counts are unavailable. They are unavailable. Substituting the fallback:

$$\text{underservice} = (\text{need} \times \text{pop}) / (\text{pop} / \text{norm}) = \text{need} \times \text{norm}$$

The population term cancels **exactly**. A supply-adjusted score would be the need index times a terrain constant — carrying no information about existing infrastructure while appearing to account for it, and ranking districts almost identically to raw need, so the error would never look suspicious. A database view proves this on the real data for all 696 districts, and a test enforces it. No supply-adjusted figure is published.

2 · Optimisation did not beat sorting — and that is the result

An earlier version of this project claimed a **+13% optimisation premium**. It was wrong. The premium was measured against a greedy baseline sorted by the *need index* while the objective being maximised was *coverage gain* — a straw man. Sorted by the quantity it is supposed to maximise, greedy reaches the optimum exactly, choosing the identical 25 districts. Verified across 11 constraint configurations.

That is a property of the problem, not a failure of the solver: the objective is linear and the per-state caps form a partition matroid, which is precisely the structure greedy handles optimally. So what does the integer program buy?

A proof	Greedy gives an answer; the solver certifies it cannot be beaten. Without it we would not have known — and at a region floor of 2 or more, our greedy was silently returning infeasible answers that scored higher than the optimum
Infeasibility detection	40 facilities at one per state across ~30 states is impossible. The solver says so; a heuristic returns a plausible-looking list
Declarative constraints	Changing the cap is one line of config. The greedy repair logic had to be rewritten to handle a region floor above 1
Robustness	Once constraints couple districts — adjacency, travel time, shared staffing — the matroid structure is gone and greedy loses its guarantee. The formulation does not change

3 · The first objective was quietly discriminatory

The original objective produced an allocation containing **zero hilly and zero tribal districts** — though 23% of candidates are non-plains. The cause was subtle: the cap in *need* × *min(catchment_norm, rural_population)* binds for only 2 of 696 districts, so the objective collapsed to need × 5,000 in plains and need × 3,000 in hills. A hilly district needed a 67% higher need index merely to compete.

> THE ERROR
>
> This inverts the purpose of the standard it was built on. The IPHS norm is *lower* in hilly and tribal areas precisely **because** those areas are harder to reach. The model was penalising hard terrain for being hard to serve.

The default objective is now terrain-neutral: one facility counts as one facility wherever it stands. Both are computed and the choice is explicit in configuration, because it is a policy judgement rather than a technical one. It moves 9 of the 25 districts, and the final allocation contains 5 hilly and 4 tribal districts.

4 · Some districts are far sicker than their poverty explains

The need index says *how bad* outcomes are. It cannot separate a district that is unhealthy **because it is poor** from one that is unhealthy **beyond what its poverty explains** — and a new facility plausibly helps the second far more. So the index was modelled from Census socioeconomic variables and the **residuals** studied. Out-of-fold **R^2 = 0.657** (random forest, 5-fold; ridge 0.537).

A high R^2 would *not* be a good result. It would mean health outcomes are fully determined by material conditions, leaving no room for a health-system signal. The useful quantity is what the model cannot explain.

District	State	Actual	Predicted	Residual
Kamrup Metropolitan	Assam	0.622	0.171	+0.451
Mewat	Haryana	0.835	0.461	+0.374
Jalgaon	Maharashtra	0.724	0.380	+0.345
Deoghar	Jharkhand	0.919	0.584	+0.335
Patna	Bihar	0.810	0.536	+0.274

Kamrup Metropolitan is Guwahati — a state capital with a strong socioeconomic profile and health outcomes far below it. Mewat is a documented outlier despite sitting an hour from Delhi. Patna is another state capital. A need ranking alone surfaces none of them. The residual is **not** used to drive the allocation; folding it in without a causal argument would be indefensible.

How the answer was stress-tested

The weight vector is the least defensible choice in the whole methodology. It encodes a value judgement — whether child stunting matters as much as institutional births — and no amount of data settles it. So rather than defend one vector, we measured how much the answer depends on it: 10,000 sampled weightings per regime, with the full optimisation re-solved 200 times each.

Regime	What it represents	Robust
Centred	Plausible committee disagreement	11 of 25

Regime	What it represents	Robust
Uniform	Adversarial: allows 80% weight on a single indicator	2 of 25

Both numbers are reported. Quoting only the flattering one would not survive scrutiny.

A subtlety worth naming: judging an allocated district by whether it stays in the national top 25 is the *wrong test*. The constrained optimum deliberately reaches outside that list, because the cap forbids taking 12 districts from Bihar and the floor requires every region. Districts are therefore judged on whether they survive a full re-solve — which mislabels nothing.

Engineering, not just analysis

76 automated tests	Covering crosswalk integrity, database constraints, index mathematics, optimiser feasibility and dependency drift
Reproducible by construction	Every source is pinned and checksummed; the whole pipeline rebuilds with one command; the sensitivity seed is fixed so published figures are verifiable
Constraints in the schema	The database refuses to hold a contradiction — an indicator cannot enter the index without a direction and a domain; a split flag cannot drift from its sibling count
Fails loudly	A missing configuration parameter raises rather than returning NULL. That exact bug once reported an allocation with 12 facilities in one state as violating nothing
Runs without a database	The app falls back to an exported snapshot and says so on screen, which is what makes cloud deployment possible

Validation that is not self-referential

Total population across all 705 reconciled districts was checked against the published Census total:

Loaded across 705 districts	1,209,735,047
Census of India 2011, all-India	1,210,854,977

Gap	1,119,930
Chandigarh + Lakshadweep (no NFHS factsheet)	1,119,923
Unexplained residual	7 people

Population is conserved to seven people across 118 districts whose population had to be apportioned. Had the split logic been double-counting, this number would be in the millions. It is now a test, so it cannot regress quietly.

The failure worth knowing about

Fuzzy matching district names did not fail loudly — it failed *plausibly*. Districts created after 2011 differ from an existing district by a single directional word, and the string matcher scored those pairs 80-90, comfortably inside any sane acceptance threshold:

District	Wrongly matched to	Score
South East (Delhi)	North East	80
North Garo Hills	South Garo Hills	88
South West Khasi Hills	West Khasi Hills	84

Each would have given a district its **sibling's** population — and the output would have looked perfect, because a high match score is exactly what you check for. Raising the threshold does not help: the bad match scores 80 while the genuine rename Darjeeling to Darjiling scores 84. The fix was a directional-token guard that refuses a match outright when the directional words disagree, forcing 76 districts into a hand-built override table.

Questions to expect, and how to answer them

“How did you join datasets with mismatched keys?”

State-restricted fuzzy matching, because a national match confuses the Aurangabad in Maharashtra with the one in Bihar. Then a directional-token guard, because post-2011 splits score 80-90 against their own siblings and get silently accepted. Then a hand-built override table of 76 districts. Validated against the published Census population total — it reconciles to 7 people.

“Why those weights?”

I don't defend them. I sampled 10,000 alternatives and report that 11 of 25 districts survive any plausible re-weighting while 14 depend on the choice — and I name which. Under an adversarial regime only 2 survive, and I report that too.

“What did optimisation buy you over sorting?”

Not a bigger number — I checked, and a correct greedy matches the optimum on this problem because the constraints form a partition matroid. It buys a proof, infeasibility detection, and constraints I can change without rewriting the selection logic. I originally claimed a 13% premium, found it was measured against a straw man, and withdrew it.

“What's the weakest part?”

There is no supply data. This measures need, not unmet need. And when I substituted the norm-implied supply the methodology allows, the adjustment cancelled algebraically — so I publish no supply-adjusted figure at all.

“Did anything surprise you?”

The first objective produced 25 plains districts and zero hilly or tribal ones. It looked like a modelling detail and was actually a fairness failure: it penalised hard terrain for being hard to serve, which inverts the standard it was built on.

For a CV

Long form

Built an end-to-end analytics system allocating public health infrastructure across 705 Indian districts: reconciled NFHS-5 (2019-21) survey data with Census 2011 boundaries across a decade of district splits (100% match rate, population conserved to 7 people against the published national total), modelled a composite need index in PostgreSQL, and solved a budget-constrained integer program in PuLP/CBC. Stress-tested the result against 10,000 Dirichlet-sampled weightings to separate robust from contested recommendations. 76 automated tests; interactive Streamlit tool deployed publicly.

Bullet form

- Reconciled two Indian government datasets a decade apart — 705 districts, 100% matched — building a fuzzy crosswalk with a directional-token guard that caught silent sibling mismatches a naive threshold accepted; validated by conserving population to 7 people against the published Census total.
- Modelled a 7-indicator composite need index as a chain of PostgreSQL materialised views with schema constraints encoding the methodology, so the database cannot hold a contradiction.
- Formulated and solved facility allocation as an integer program (PuLP/CBC) under equity constraints; proved a correct greedy attains the same optimum and reported the honest implication rather than a headline premium.
- Quantified model uncertainty with 10,000 Dirichlet-sampled weight vectors and 400 optimiser re-solves, classifying 11 of 25 recommendations as robust and 14 as contested.
- Identified and corrected a fairness defect where the objective excluded all hilly and tribal districts by penalising the terrain-adjusted service norm.
- Shipped an interactive Streamlit application with live re-optimisation and a district choropleth, backed by 76 automated tests and a one-command reproducible pipeline.

Limitations

Stated up front, because volunteering them is stronger than being caught by them.

No supply data	District-level facility counts are not published in machine-readable form. This measures need, not unmet need — the single largest weakness
Decade-old population	Census 2011 against 2019-21 outcomes, with no projection, because no defensible district-level growth assumption exists

Derived rural population	Computed from rural household share; understates rural population by about 1.1%, which is conservative for this purpose
Apportioned splits	118 districts share a Census parent; population divided equally — unbiased in aggregate, wrong district by district
Third-party extraction	The survey data was parsed from published PDFs by a third party who documents possible errors
Narrow indicator set	Seven maternal, child and nutritional indicators. Nothing on communicable disease, mental health or injury — and no amount of weight sampling fixes a shared blind spot
No geography in the objective	No travel time, no existing facility locations, no construction cost

This is a portfolio project built on public data. It is not a policy recommendation and should not be presented as one.